

MOTTEC, Switzerland

WMO: 067160

Lat: 46.1478N

Lon: 7.6242E

Elev: 1580

StdP: 83.73

Time Zone: 1.00 (EUC)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -12.1	(c) -10.0	(d) -17.4	(e) 1.0	(f) -9.5	(g) -15.3	(h) 1.2	(i) -6.8	(j) 9.0	(k) 2.6	(l) 8.0	(m) 1.7	(n) 2.6	(o) 180	(p) 0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 9.7	(c) 28.9	(d) 17.6	(e) 26.9	(f) 16.9	(g) 25.2	(h) 16.3	(i) 18.6	(j) 26.7	(k) 17.7	(l) 25.3	(m) 16.8	(n) 23.8	(o) 5.3	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	13.5	22.2	14.9	12.9	21.1	14.1	12.2	20.5	59.7	26.9	56.5	25.4	53.7	23.9	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.6	9.0	7.6	-12.7	30.3	4.8	3.4	-16.1	32.8	-18.9	34.8	-21.6	36.7	-25.1	39.2
			-13.9	18.7	4.3	2.0	-17.0	20.2	-19.5	21.3	-21.9	22.5	-25.1	23.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-1.2	0.1	4.7	7.7	11.4	16.2	17.6	17.8	13.1	9.1	3.6	0.2		
	DBStd	8.01	4.48	5.01	5.20	4.53	4.99	4.68	3.83	4.23	4.12	4.17	4.44	3.99		
	HDD10.0	1531	346	276	176	97	44	4	1	2	17	68	195	304		
	HDD18.3	3754	604	509	423	320	221	98	61	62	164	287	443	562		
	CDD10.0	946	0	1	11	27	87	188	237	243	109	39	2	1		
	CDD18.3	127	0	0	0	0	5	33	38	45	6	0	0	0		
	CDH23.3	831	0	0	0	1	31	241	246	298	14	0	0	0		
	CDH26.7	193	0	0	0	0	2	59	50	82	0	0	0	0		
(14) Wind	WSAvg	2.6	2.1	2.6	2.9	3.1	3.1	3.0	2.9	2.8	2.6	2.2	2.1	2.2		
(15) Precipitation	PrecAvg	706	51	48	48	49	71	67	67	68	51	56	67	62		
	PrecMax	920	147	205	113	98	161	130	154	143	155	167	186	158		
	PrecMin	439	10	2	14	3	23	30	21	26	14	5	17	5		
	PrecStd	115	35	46	26	26	42	27	26	29	32	37	43	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	12.8	18.3	21.8	26.4	30.7	30.5	32.6	25.2	20.9	14.0	11.4		
		MCWB	5.0	5.6	8.4	11.0	15.1	18.1	17.4	17.7	16.4	12.8	7.7	5.5		
	2%	DB	7.5	10.2	15.6	18.9	23.7	28.6	27.9	28.7	23.2	18.2	12.2	8.0		
		MCWB	3.3	3.7	7.7	9.6	13.8	17.3	17.5	17.5	15.5	11.2	6.7	3.9		
	5%	DB	6.0	8.2	13.6	16.5	21.2	26.3	26.0	26.4	21.0	16.7	10.4	6.4		
		MCWB	2.7	3.1	6.7	8.3	12.5	16.4	16.9	16.9	14.1	10.6	5.8	3.2		
	10%	DB	4.3	6.5	11.6	14.6	19.0	24.0	24.2	24.5	19.2	14.9	9.0	5.2		
		MCWB	1.5	2.4	5.6	7.4	11.5	15.4	15.9	16.3	13.2	9.7	5.6	2.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	6.3	9.7	11.8	16.1	19.3	19.4	19.6	17.5	13.6	9.8	6.5		
		MCDB	7.4	9.8	15.3	19.1	24.4	28.6	27.4	28.6	23.9	18.2	11.5	9.0		
	2%	WB	4.1	5.0	8.5	10.1	14.5	18.3	18.4	18.4	15.9	12.5	8.2	5.4		
		MCDB	6.7	8.7	14.0	17.7	21.8	27.0	26.0	26.4	21.5	17.2	10.2	7.6		
	5%	WB	3.0	3.8	7.4	9.0	13.2	17.1	17.5	17.5	14.9	11.4	6.7	3.9		
		MCDB	5.3	7.5	12.4	15.6	19.9	24.4	24.8	24.9	19.6	15.3	9.4	6.0		
	10%	WB	1.8	2.7	6.2	7.8	12.0	16.0	16.6	16.6	14.0	10.3	5.8	2.5		
		MCDB	4.0	6.1	11.1	13.7	18.2	22.9	23.1	23.4	18.1	13.9	8.4	4.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.9	8.6	9.4	9.5	9.9	10.2	9.7	8.4	7.6	5.8	5.5		
		MCDBR	5.9	8.0	10.0	11.5	11.6	12.4	11.4	11.4	9.8	9.0	7.3	6.3		
	5% WB	MCWBR	3.4	4.5	5.2	5.6	5.8	5.5	5.0	5.0	4.9	4.7	4.2	3.9		
		MCWBR	5.5	6.8	9.1	10.6	10.4	10.7	10.3	10.1	8.2	7.5	5.5	4.5		
(40) Clear-Sky Solar Irradiance	taub		0.239	0.266	0.313	0.358	0.377	0.382	0.378	0.365	0.341	0.314	0.273	0.238		
		taud	2.304	2.236	2.162	2.160	2.210	2.245	2.255	2.304	2.362	2.411	2.407	2.340		
	Ebn at Noon		879	914	909	895	888	883	881	880	873	848	829	840		
		Edh at Noon	75	100	126	140	140	138	134	131	120	103	84	69		
(44) All-Sky Solar Radiation	RadAvg		1.48	2.36	3.48	4.40	5.01	5.96	5.96	5.18	4.04	2.62	1.52	1.22		
	RadStd		0.17	0.32	0.47	0.59	0.43	0.51	0.48	0.42	0.34	0.29	0.22	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (107 neighbors)		N/A	N/A	N/A	+0.72	+0.41	+0.31	N/A	N/A	+56	+16

Nomenclature: See separate page